Introduction to Artificial Intelligence (AI)

Professor Nihad Karim Chowdhury
Department of Computer Science and Engineering
University of Chittagong

Lecture Overview

- Objective: Understand key concepts of AI, its history, applications, and the distinction between AI, Machine Learning, and Deep Learning.
- Key Topics:
  - AI Definition
  - History of AI
  - Applications of AI
  - AI vs. ML vs. DL
  - Agents & Environments
  - Ethical Considerations

Definition of AI (Part 1)

- AI is the science and engineering of making intelligent machines capable of performing tasks that require human intelligence.
- Originates from multiple disciplines: computer science, biology, psychology, linguistics, mathematics, etc.

Definition of AI (Part 2)

- Intelligence in AI:
  - Perception
  - Reasoning
  - Learning
  - Problem-solving
  - Decision-making

The Scope of AI

- Main Fields of AI:
  - Machine Learning
  - Natural Language Processing
  - Robotics
  - Computer Vision
  - Expert Systems

History of AI (Part 1)

- Early AI Development:

  - 1950s: Alan Turing's Turing Test

  - 1956: Dartmouth Conference – Start of AI as a field

  - Early AI systems: Logic Theorist and General Problem Solver (GPS)

History of AI (Part 2)

- AI Winters:
  - Periods of reduced funding and interest due to unmet expectations.
  - Key setbacks in 1970s and 1980s.

History of AI (Part 3)

- Modern AI:
  - 1990s: AI rebirth with expert systems.
  - 2000s: Machine learning and deep learning breakthroughs.
  - 2010s: AI applications in industries like healthcare, finance, and transportation.

Applications of AI (Part 1)

- Healthcare:
  - Diagnosis (IBM Watson)
  - Drug discovery
  - Personalized treatment plans

Applications of AI (Part 2)

- Transportation:
  - Self-driving cars (Tesla, Waymo)
  - AI for route optimization and logistics

Applications of AI (Part 3)

- Finance:
  - Fraud detection
  - Algorithmic trading
  - Credit risk assessment

Applications of AI (Part 4)

- Retail:
  - Personalized shopping experiences
  - Inventory management with AI algorithms

AI vs. Machine Learning (Part 1)

- AI: The broader concept of machines being able to carry out tasks that are considered "smart.“
- Machine Learning (ML): A subset of AI focused on the idea that systems can automatically learn from data and improve over time without being explicitly programmed.

AI vs. Machine Learning (Part 2)

- How ML fits into AI:
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning

Machine Learning Types (Part 1)

- Supervised Learning:
  - Algorithms learn from labeled data.
  - Common algorithms: Linear regression, decision trees.

Machine Learning Types (Part 2)

- Unsupervised Learning:
  - Algorithms identify patterns in unlabeled data.
  - Common algorithms: K-means clustering, PCA.

AI vs. Deep Learning

- Deep Learning (DL):
  - A subset of machine learning.
  - Uses neural networks with many layers (deep architectures) to learn complex patterns from large datasets.

Deep Learning Architectures

- Neural Networks:
  - Layers of interconnected "neurons.“
  - Activation functions.
- Common Applications:
  - Image recognition (CNNs).
  - Natural language processing (RNNs, Transformers).

Key Concepts in AI: Agents

- Definition:
  - An agent is an entity that perceives its environment and acts upon it.
  - Goal: Maximize performance based on perception and actions.

Components of an AI Agent

- Components of an AI Agent
  - **Perception:** The agent observes the environment, collecting data or information that reflects the current state. This data is gathered via sensors (e.g., cameras, microphones, radar).
  - **Decision-making:** Based on its perception, the agent decides on the best course of action by applying algorithms, logic, or learned behaviors.
  - **Action:** After making a decision, the agent interacts with the environment by taking actions through actuators (e.g., movement of a robot's arm, sending a message in a software system).
  - **Feedback:** The agent continuously receives feedback from its actions, adjusting its decisions to improve future performance.

Types of Agents

- **Simple Reflex Agents:** These agents select actions based on the current percept without considering the history of previous states. They work well in fully observable environments where the current state provides enough information to act.
  - **Example:** A thermostat that adjusts the temperature based on the current room temperature.
- **Model-based Reflex Agents:** These agents maintain an internal state to track parts of the world that are not visible. They use a model of the world to handle partially observable environments.
  - **Example:** A self-driving car uses internal models to keep track of the positions of other vehicles and pedestrians that may not always be visible.

Types of Agents

- **Goal-based Agents:** These agents choose actions based on a goal. In addition to knowing the current state of the world, they make decisions by considering future outcomes that help them achieve specific goals.
  - **Example:** A chess-playing AI calculates the best moves to reach the end-game position (goal) of checkmating the opponent.
- **Utility-based Agents:** These agents not only aim to achieve a goal but also try to optimize their performance by maximizing a utility function.
  - **Example:** A recommendation system that suggests products to users, not only based on preferences but also by maximizing customer satisfaction and engagement.

Example: Self-Driving Car (An AI Agent)

- A self-driving car can be considered an AI agent:
  - **Perception:** It uses sensors like cameras, LiDAR, and GPS to perceive its environment. It gathers data about the road, obstacles, traffic signals, pedestrians, and other vehicles.
  - **Decision-making:** The car's AI processes this information to make decisions about speed, direction, and when to stop or turn. It also takes into account the laws of the road and real-time traffic information.
  - **Action:** Based on its decisions, the car adjusts its speed, changes lanes, or applies brakes by controlling its actuators (throttle, steering, brakes).
  - **Feedback:** The car constantly monitors its surroundings and makes adjustments to its behavior to avoid collisions, ensure a smooth ride, and reach its destination efficiently.

Key Concepts in AI: Environments

- Definition:
  - An environment is the external context in which an agent operates.

Types of Environments in AI

- Fully Observable vs. Partially Observable Environments:
  - **Fully Observable:** The agent has complete access to the environment's state at all times. The information needed to make optimal decisions is readily available.
    - Example: A chess game, where both players can see the entire board and all pieces at all times.
  - **Partially Observable:** The agent has only limited access to information about the environment, often requiring it to make decisions based on incomplete data.
    - Example: A self-driving car, which cannot "see" all the traffic or road conditions beyond its sensors.

Types of Environments in AI

- Deterministic vs. Stochastic Environments:
  - **Deterministic:** The next state of the environment is entirely predictable, given the current state and the action taken by the agent.
    - Example: Solving a puzzle, where each move leads to a predictable outcome.
  - **Stochastic:** The environment includes some level of randomness or uncertainty, and the same action may lead to different outcomes.
    - Example: A weather prediction system where the environment has a degree of uncertainty due to many variables.

Types of Environments in AI

- Static vs. Dynamic Environments:
  - Static: The environment does not change while the agent is deliberating. The agent can take its time to decide.
    - Example: A crossword puzzle where the puzzle state remains fixed as the agent works on it.
  - Dynamic: The environment can change while the agent is making decisions or taking actions.
    - Example: A stock market trading system where prices and other factors change constantly in real-time.

Types of Environments in AI

- Discrete vs. Continuous Environments:
  - Discrete: The environment has a finite number of distinct states, and actions happen in discrete steps.
    - Example: A board game like Monopoly, where the game moves in steps.
  - Continuous: The environment has an infinite number of states, and actions occur in a continuous space.
    - Example: A robot navigating a room, where its movements and the space it can occupy are continuous.

Types of Environments in AI

- Episodic vs. Sequential Environments:
  - Episodic: Each action taken by the agent is independent of previous actions. The agent's experience is divided into discrete episodes.
    - Example: Image recognition tasks, where each image is processed independently of others.
  - Sequential: Each action has consequences that affect future decisions. The agent must consider the long-term effects of its actions.
    - Example: A chess game, where each move affects the future state of the game.

Example: Self-Driving Car Environment

- A self-driving car operates in a highly dynamic, partially observable, and stochastic environment:
  - **Partially Observable:** The car uses sensors like cameras, radar, and LiDAR to perceive its surroundings, but it cannot observe all aspects of the environment (e.g., objects hidden behind other cars or buildings).
  - **Dynamic:** The environment is constantly changing with other cars, pedestrians, and weather conditions affecting the car's behavior.
  - **Stochastic:** Uncertainty is present in the form of unpredictable behavior from other drivers, pedestrians, and changing weather.
  - **Continuous:** The car moves in a continuous space, adjusting its speed, steering, and braking fluidly to navigate the environment.
  - **Sequential:** Each decision (e.g., accelerating or turning) has long-term consequences for reaching the destination safely and efficiently.

State Space in AI



State:

A representation of the environment at a given time.



State Space:

The set of all possible states in an environment.

State Space in AI

- In AI, a **state** refers to a specific condition or configuration of an agent or system at any given moment. It represents the current situation that the agent is experiencing or the current stage of the problem it is trying to solve. States are crucial in defining the environment and determining the actions an agent can take. States can be discrete (well-defined) or continuous (changing smoothly over time).
- The **state space** is the collection of all possible states an agent can be in, which is often explored during the problem-solving or decision-making process. Each state usually includes all relevant information about the environment, agent's position, and any other variables needed to determine the next action.

Example: State in a Maze Navigation Problem

- Imagine an AI agent is tasked with solving a maze. In this example, a state refers to the agent's position in the maze at any given moment. Let's break it down:

1. **State:** Each state in the maze is defined by the agent's current position (x, y) within the grid. For example, in a 5x5 grid maze:
  - Initial State: (0, 0) - The agent starts at the top-left corner.
  - Goal State: (4, 4) - The agent needs to reach the bottom-right corner.

Example: State in a Maze Navigation Problem

2. **Actions:** The agent can take four actions at each state:

Move up.

Move down.

Move left.

Move right.

Each action will change the state of the agent by updating its position.

3. **State Transition:** When the agent moves, it transitions from one state to another. If the agent is at (1, 1) and moves right, its new state becomes (1, 2).

4. **State Space:** The state space for this problem includes all the possible locations (x, y) the agent can be in, ranging from (0, 0) to (4, 4).

Example of State Transitions in a Maze:

Initial State: (0, 0) Agent starts at the top-left corner.

Action: Move right The agent moves from (0, 0) to (0, 1).

New State: (0, 1)The agent is now at position (0, 1).

Action: Move down The agent moves from (0, 1) to (1, 1).

New State: (1, 1)The agent continues exploring the maze from this new position.

Pictorial Example of AI Agent, Environment, and State

Let's consider an AI Agent in the context of a self-driving car in a city environment. This example will demonstrate the relationship between the agent, its environment, and the concept of state.

Pictorial Example of AI Agent, Environment, and State

1. **AI Agent: Self-Driving Car**

   The AI agent in this scenario is a self-driving car equipped with sensors (cameras, LiDAR, radar, etc.) and actuators (controls for steering, acceleration, braking). The car's AI system processes data from its environment and makes decisions such as moving forward, stopping, turning, or changing speed.

Pictorial Example of AI Agent, Environment, and State

2. **Environment: City Street**

The environment in this case is a city street. The environment includes:

Roads: The car moves on them.

Traffic signals: Red, yellow, and green lights.

Pedestrians: People crossing the street.

Other vehicles: Cars, trucks, buses, and bikes.

Obstacles: Construction barriers, potholes, or any obstruction in the road.

Pictorial Example of AI Agent, Environment, and State

3. State: The Car's Situation

The state refers to the specific condition of the agent at any point in time. In the case of the self-driving car, the state includes:

Position: The car’s current GPS location.

Speed: The car’s current velocity.

Traffic conditions: The status of nearby vehicles, traffic signals, and pedestrians.

Sensor data: Information from cameras, LiDAR, and radar about the surrounding environment.

Example of AI Agent, Environment, and State

State example:

- **Position:** (52.376552, 4.897070) – The car is on a street corner.
- **Speed:** 35 km/h.
- **Traffic signal:** Green light.
- **Nearby vehicle:** A car is 10 meters ahead.
- **Pedestrians:** No pedestrians in the crosswalk.

Example Visualization

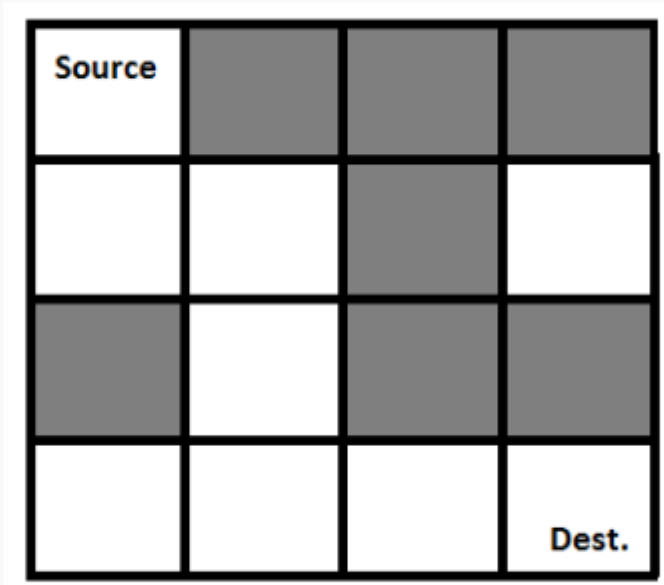
Environment	Agent	State
?	?	?

Example Visualization

Environment	Agent	State
A busy city street with moving traffic, traffic signals, pedestrians, and obstacles.	A self-driving car equipped with sensors and AI algorithms.	Current location: Main St. intersection, Speed: 35 km/h, Traffic signal: Green, Nearby vehicles: 1 car 10 meters ahead, No pedestrians in crosswalk.

Challenge: Rat in Maze

Input:



Output: *DRDDRR*

Challenge : Maze Navigation

- **Objective:**
  - Students will design an AI agent to navigate a maze from a starting point to a goal while avoiding obstacles.

Challenge : Maze Navigation

- **Instructions:**
  - Provide students with a grid (5x5 or 10x10) where certain cells are obstacles, and some are open paths.
  - Define the start point (e.g., top-left corner) and the goal (e.g., bottom-right corner).
  - The agent (a simple algorithm) needs to find the shortest or most efficient path to the goal.
- **Steps:**
  - Design the maze as a grid where 1 represents open space and 0 represents obstacles.
  - Have students write code to simulate the AI agent that can move up, down, left, or right.
  - The AI agent updates its state (position) after every move and tries to avoid obstacles while finding the goal.

Explanation in the context of a 4x4 Maze

Environment	Agent	State
?	?	?

Explanation in the context of a 4x4 Maze

Environment	Agent	State
The environment is the maze itself, represented as a 2D grid (e.g., 4x4 or 5x5). Some grid cells are open spaces (where the agent can move), and some are obstacles (where the agent cannot enter). The environment defines the rules for how the agent interacts with the maze. The boundaries of the maze are also part of the environment, restricting the agent's movement.	The AI agent is a virtual entity that can move in the maze. Its task is to navigate from the start to the goal. The agent's actions include moving in four possible directions: up, down, left, and right. Each move changes its state (position) in the maze.	The state represents the agent's current position in the maze. At each step, the state is defined by the coordinates (x, y) of the agent in the grid. For example, if the agent is at the top-left corner of a 4x4 grid, its state would be (0, 0). The agent's state updates every time it takes an action (move). The goal state is the position where the agent reaches the bottom-right corner (e.g., (3, 3) in a 4x4 grid).

Challenge: Chess Piece Movement (Agent in a Board Game)

- **Objective:**
  - Simulate an AI agent (chess piece) that moves across a chessboard following the rules of the game, with the goal of capturing a specific piece.

Challenge: Chess Piece Movement (Agent in a Board Game)

- **Instructions:**
  - Define a simplified 8x8 chessboard with a few chess pieces.
  - The AI agent (e.g., a queen) must navigate the board according to the rules and attempt to capture an opponent's piece.
  - The environment is the chessboard, and the state is the position of all pieces on the board.
- **Steps:**
  - Set up an 8x8 board with a few chess pieces (e.g., a queen and a pawn).
  - Have students write code that simulates the movement of the queen, following the chess rules.
  - The AI agent’s state is updated after each move (e.g., new position of the queen).

Explanation in the context of the chessboard

Environment	Agent	State
?	?	?

Explanation in the context of the chessboard

Environment	Agent	State
The environment is the 8x8 chessboard with pieces placed on it. The environment includes: pieces, rules of movement for each piece, and the objective of capturing an opponent's piece. The environment also imposes boundaries (the edges of the board) and constraints (other pieces) that restrict where the agent can move.	The AI agent is a chess piece (e.g., Queen) that can move according to the rules of chess. The agent's goal is to capture an opponent's piece by selecting optimal moves. The agent's actions include valid moves (e.g., diagonal, horizontal, vertical for the queen) on the chessboard.	The state represents the current configuration of the board, including the position of all pieces (both friendly and opponent pieces). The agent's state is updated after each move

Ethical Considerations in AI (Part 1)

- Fairness:
  - AI systems must avoid biases that can lead to unfair treatment in critical areas such as hiring, loan approvals, and policing.

Ethical Considerations in AI (Part 2)

- Transparency:
  - AI systems must be transparent about their decision-making processes, particularly when deployed in sensitive applications.

Ethical Considerations in AI (Part 3)

- Accountability:
  - Developers and users of AI must be held accountable for the actions of AI systems.

Ethical Considerations in AI (Part 4)

- Privacy:
  - AI systems must respect individuals' privacy by handling data responsibly and securely.

Challenges in AI Ethics

- Bias in AI systems:

  - Pre-existing biases in data leading to unfair outcomes.

- Job displacement:

  - Impact on employment due to automation.

Self Study

- Rationality in AI

Conclusion

- AI is a transformative field impacting many areas of life.
- Understanding its foundations, ethical implications, and relation to ML and DL is crucial for anyone working in technology today.

Book Reference

- Stuart Russell and Peter Norvig: “Artificial Intelligence a Modern Approach”, 3rd Edition, Pearson Education Asia. (Text Book) (Online) (Lectures) Chapter 1 and 2.